

Mid Exam

Sukkur IBA University

Department of Computer Science

Course: Advanced Natural Language Processing (MS/PhD CS/AI/SE)

Midterm Examination – Practical (Machine Based, Open Book)

Duration: 120 Minutes

Total Marks: 100

Submission Instruction:

Please submit your Google Colab notebook (ipynb) via LMS. You should download it on your machine and then upload on the LMS link

You can use any material, ChatGPT, Claude or any other LLM chat, refer books, course material etc.

During exam, you are not allowed to use any communication with peers or outside world (no whatsapp, email, phone etc.) Immediate disqualification if you are found using any of these during the exam.

The dataset to be used with all four tasks is downloadable from the course eLearning with this exam.

Task 1 — Load and Explore the Dataset (10 Marks)

Load and analyze the dataset.

Requirements

Load 1600 samples from the AG News training set, 400 samples from each class labels. The dataset is provided on eLearning with the mid exam

Replace numeric labels to word labels ("world", "sports", "business", and "science" for 1, 2, 3 and 4 respectively)

Convert them to a pandas DataFrame.

Display the first 5 rows.

Print the dataset shape.

Show the distribution of categories. This should be using a bar graph with each bar having word class label, also this should be on full dataset, NOT on the 1600 samples. This is just to see distribution of different classes in original dataset.

Task 2 — Text Embedding Generation (20 Marks)

Convert news articles into vector embeddings.

Requirements

Use SentenceTransformers.

Load the model: sentence-transformers/all-MiniLM-L6-v2 (or anyother model of your choice)

Generate embeddings for all 1600 samples.

Print the shape of the embedding matrix.

Expected Output

Embedding matrix shape: (1600, 384)

Task 3 — Implement Semantic Search (30 Marks)

Build a semantic search system.

Requirements

Create a function:

```
def semantic_search(query, k=3):
```

Steps:

Convert query into embedding

Compute cosine similarity

Return top-k most relevant news articles

Display:

Article (sample) text (first 150 characters)

predicted category (remember word categories are "world", "sports", "business", and "Science") ,you need to write word categories, not the numbers.

similarity score

Example:

Test Query

"latest developments in artificial intelligence"

Example Output:

Top Results:

1. Science | score: 0.83

Researchers unveiled a new AI system capable of...

2. Science | score: 0.80

A major technology company announced progress...

3. Business | score: 0.76

Tech firms continue to invest heavily in AI...

Task 4 — Build a Simple Retrieval-Augmented Generation (RAG) System (40 Marks)

Use retrieved documents to generate answers using an LLM.

Steps

Use your semantic search function to retrieve top 3 documents.

Construct the following prompt:

Use the context below to answer the question.

Context:

{retrieved articles}

Question:

{query}

Provide a concise answer.

Generate an answer using a HuggingFace model.

Remember, retrieved articles are coming from RAG, you can use RAG vector database of your choice.

Suggested model:

google/flan-t5-base or any other model of your choice for LLM

Example Query

"What are the recent developments in AI technology?"

Expected Output

A short generated answer using the retrieved context.

AG_News_Corpus.csv

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